📊 FINAL PROJECT REPORT - MARKETING ROI DASHBOARD

📋 EXECUTIVE SUMMARY

Metric	Value	Business Impact
Total Campaigns Analyzed	7,500+	Comprehensive analysis
Time Period	Jan 2025 - Feb 2026	14 months data
Total Marketing Spend	$5.2 Million	Budget under review
Total Revenue Generated	$30.4 Million	5.8x ROI
Total Profit	$12.4 Million	Bottom line impact
Wasted Spend Identified	$420,000	8% of budget
Optimization Potential	$1.17 Million	22.5% improvement

PROJECT ARCHITECTURE

Phase	Tool Used	Purpose	Key Activities
Data Storage	AWS S3	Cloud Storage	Bucket: `marketing-roi-analytics-2026`
Data Querying	AWS Athena	Serverless SQL	Database: `marketing_analytics`
Data Cleaning	Python	Data Preparation	10 lines of code
Visualization	Power BI	Interactive Dashboard	3 Pages
Machine Learning	Power BI Auto ML	Key Influencers & Forecast	Zero Code

PHASE 1: PYTHON DATA CLEANING

Step	Action	Before	After	Business Impact

📊 FINAL PROJECT REPORT - MARKETING ROI DASHBOARD

| 1 | Load Raw Data | 7,500 rows, 16 columns | 7,500 rows, 28 columns | More features for analysis |

| 2 | Clean Cost Column | `"$16,174.00"` | `16174.0` | Accurate ROI calculation |

| 3 | Date Processing | String format | DateTime + Year/Month/Quarter | Time intelligence ready |

| 4 | Create Business Metrics | Only ROI | CTR, CPC, Profit, ROAS | 5x more insights |

| 5 | Category Standardization | Raw values | Channel Groups, Gender Target | Executive-friendly view |

| 6 | ROI Segmentation | None | Poor/Good/Excellent tiers | Easy prioritization |

| 7 | Outlier Detection | Hidden | Flagged (1% of data) | Identify anomalies |

| 8 | Export Cleaned Data | Raw CSV | `marketing_roi_cleaned.csv` | Ready for Power BI |

☁️ PHASE 2: AWS STORAGE + ATHENA

| Component | Name | Details | Purpose |

|----------|------|---------|---------|

| S3 Bucket | `marketing-roi-analytics-2026` | Region: ap-south-1 | Data lake for raw/cleaned data |

| Athena Database | `marketing_analytics` | Serverless SQL | Query without loading |

| Athena Table | `marketing_roi` | 28 columns | External table on S3 |

| Key SQL Queries| | | |

| | Channel Performance | `SELECT channel_used, AVG(roi) ...` | ROI by channel |

| | Wasted Spend | `SELECT SUM(cost) WHERE roi < 3` | $420K identified |

| | Top Campaigns | `ORDER BY profit DESC LIMIT 10` | Best performers |

| | Monthly Trend | `GROUP BY year, month` | Seasonal patterns |

📊 PHASE 3: POWER BI DASHBOARD - PAGE 1 (Executive Dashboard)

📊 FINAL PROJECT REPORT - MARKETING ROI DASHBOARD

| Visual Type | Columns Used | Measures | Key Insight |

|----------|----------|----------|----------|

| KPI Card 1| `impressions` | `Total Impressions = 1,245K` | ▲ 21.5% YoY |

| KPI Card 2| `clicks` | `Total Clicks = 185K` | ▲ 10.2% YoY |

| KPI Card 3 | `conversion_rate` | `Total Conversions = 12.4K` | ▲ 77.3% YoY |

| Table| `campaign_id`, `channel_used`, `roi`, `profit`, `acquisition_cost` | Top 10 by Profit | YouTube dominates |

| KPI Card 4 | `acquisition_cost` | `Total Spend = ₹52.3L` | Budget overview |

| KPI Card 5| `roi` | `Avg ROI = 5.8x` | Healthy returns |

| KPI Card 6| `profit` | `Total Profit = ₹124.5L` | Bottom line |

| Bar Chart| `channel_used`, `roi` | Avg ROI by Channel | YouTube: 7.2x (Best) |

| Line Chart| `date`, `roi` | Monthly ROI Trend | Peak in Q4 |

| Slicers | `channel_used`, `campaign_type`, `year` | Interactive Filters | Dynamic analysis |

📈 PHASE 3: POWER BI DASHBOARD - PAGE 2 (Deep Dive Analysis)

| Visual Type | Columns Used | Measures | Key Finding |

|----------|----------|----------|----------|

| Matrix | `channel_used`, `roi`, `profit`, `cost`, `clicks` | Channel Performance | YouTube: 7.2x, 1,500 campaigns |

| Stacked Bar| `target_audience`, `gender_target`, `roi` | Audience ROI | Women 35-44: 7.8x (+35%) |

| Bar Chart | `location`, `roi` | Geographic ROI | Houston: 6.7x (Best) |

| Donut Chart | `campaign_type`, `acquisition_cost` | Spend Distribution | Search: 35%, Social: 28% |

|Histogram| `roi` | ROI Distribution | Most campaigns 4-6x range |

| Bar Chart | `duration`, `roi` | Duration Impact | 30 days: 7.1x (Best) |

🤖 PHASE 3: POWER BI AUTO ML - PAGE 3 (ML Insights)

| ML Feature | Inputs | Output | Business Decision |

📊 FINAL PROJECT REPORT - MARKETING ROI DASHBOARD

|----------|--------|--------|----------|

| Key Influencers| `roi` analyzed by `channel_used`, `target_audience`, `duration`, `location` | Top factors increasing ROI | Focus on YouTube + Women 35-44 |

| | | YouTube: +1.4x impact | Shift budget to YouTube |

| | | Women 35-44: +35% | Target this segment |

| | | 30-day campaigns: +23% | Change duration strategy |

| | | Houston: +15% | Geographic focus |

| Decomposition Tree | `channel_used` → `target_audience` → `campaign_type` → `location` | Best path: YouTube → Women 35-44 → Video → Houston = 8.7x ROI | Replicate this combination |

| Forecast | Historical `roi` by `date` | Next 30 days ROI: 6.2x (±0.5) | Plan budget accordingly |

| What-If Analysis| `Budget Shift Amount` parameter | $150K shift → +$382K profit | Data-driven budget decisions |

|Recommendation Cards | ML insights + calculations | 5 actionable recommendations | Clear next steps |

💰 KEY BUSINESS INSIGHTS (TABULAR)**

| Insight Category | Finding | Value | Action |

|----------|---------|-------|--------|

| Channel Performance | YouTube | 7.2x ROI | Increase budget by 20% |

| | Google Ads | 5.8x ROI | Maintain current |

| | Website | 4.6x ROI | Reduce budget |

| Audience Performance| Women 35-44 | 7.8x ROI | Focus 30% more budget |

| | Men 25-34 | 5.2x ROI | Optimize messaging |

| | All Ages | 4.2x ROI | Reduce spend |

| Geographic Performance| Houston | 6.7x ROI | Increase by 15% |

| | Mumbai | 6.2x ROI | Maintain |

| | New York | 4.8x ROI | Review strategy |

| Campaign Duration | 30 days | 7.1x ROI | Convert all to 30-day |

| | 45 days | 5.8x ROI | Monitor |

| | 60 days | 4.9x ROI | Phase out |

| Campaign Type | Video | 7.2x ROI | Increase investment |

📊 FINAL PROJECT REPORT - MARKETING ROI DASHBOARD

| | Search | 5.8x ROI | Maintain |

| | Display | 4.5x ROI | Reduce |

💸 WASTE ANALYSIS

| Category | Campaigns | Spend | ROI | Status |

|----------|----------|-------|-----|--------|

| Low ROI (<3x) | 150 | $420,000 | 2.1x | 🔴 Cut Immediately |

| Average ROI (3-5x) | 3,200 | $2.1M | 4.2x | 🟡 Optimize |

| Good ROI (5-7x)| 2,800 | $1.8M | 6.1x | 🟢 Maintain |

| Excellent (>7x) | 1,350 | $890,000 | 7.8x | 🔵 Increase |

🎯 5 ACTIONABLE RECOMMENDATIONS

| # | Recommendation | Rationale | Expected Impact | Timeline |

|---|----------|----------|----------|----------|

| 1 | Shift $150K from Website to YouTube** | YouTube ROI (7.2x) vs Website (4.6x) | +$382,000 profit | Month 1 |

| 2 | Cut 150 low-ROI campaigns | ROI < 3x, wasting $420K | +$420,000 savings | Month 1 |

| 3 | **Focus 30% more budget on Women 35-44** | 35% higher conversion than average | +$275,000 profit | Quarter 1 |

| 4 Convert all campaigns to 30-day duration | 23% higher ROI than 60-day | +$200,000 profit | Quarter 2 |

| 5 |Increase Houston/Miami budget by 20% | 15% higher ROI than other cities | +$160,000 profit | Quarter 2 |

| | | TOTAL ANNUAL IMPACT| $1,437,000 | |

📊 TECHNICAL SPECIFICATIONS

📊 FINAL PROJECT REPORT - MARKETING ROI DASHBOARD

Component	Specification	Details
Dataset Size	7,500 rows	28 columns after cleaning
File Size	4.2 MB	Compressed CSV
Power BI Pages	3	Executive, Deep Dive, ML
DAX Measures	25+	Custom calculations
Visuals	18	Cards, Charts, Tables, ML
Slicers	5	Interactive filters
ML Features	2	Key Influencers, Forecast
AWS Services	2	S3, Athena
SQL Queries	10	Channel, Audience, Trend analysis

✅ PROJECT ACHIEVEMENTS

Achievement	Metric	Business Value
Data Volume	7,500+ campaigns processed	Comprehensive analysis
ROI Improvement Potential	22.5%	$1.17M additional profit
Waste Identified	$420,000	Immediate savings
Best Channel Identified	YouTube: 7.2x	24% above average
Best Audience	Women 35-44	35% higher conversion
Best Duration	30 days	23% better than 60-day
Best Location	Houston	15% higher ROI
ML Insights	5 key drivers	Data-driven decisions
Actionable Recommendations	5	Clear implementation plan

📊 FINAL PROJECT REPORT - MARKETING ROI DASHBOARD

📁 **DELIVERABLES**

File Name	Description	Format
`marketing_roi_cleaned.csv`	Cleaned dataset after Python	CSV
`marketing_roi_sample.csv`	1000-row sample for GitHub	CSV
`Marketing_ROI_Dashboard.pbix`	Complete Power BI dashboard	PBIX
`Marketing_ROI_Report.pdf`	Final report	PDF
`Page1_Executive.png`	Screenshot - Page 1	PNG
`Page2_DeepDive.png`	Screenshot - Page 2	PNG
`Page3_MLInsights.png`	Screenshot - Page 3	PNG
`README.md`	Project documentation	MD

🎓 SKILLS DEMONSTRATED

Skill	Tool	Proficiency Level
Data Cleaning	Python (Pandas)	Advanced
Cloud Storage	AWS S3	Intermediate
Serverless SQL	AWS Athena	Intermediate
Data Visualization	Power BI	Advanced
DAX Formulas	Power BI	Advanced
Auto ML	Power BI Key Influencers	Advanced
Forecasting	Power BI Analytics	Intermediate
What-If Analysis	Power BI Parameters	Advanced
Business Storytelling	Dashboard Design	Advanced
Presentation	Report Writing	Advanced

📊 FINAL PROJECT REPORT - MARKETING ROI DASHBOARD

🚀 NEXT STEPS FOR BUSINESS

Timeline	Action	Owner
Week 1	Cut 150 low-ROI campaigns	Marketing Manager
Week 2	Shift $150K from Website to YouTube	Finance Team
Week 3	Launch Women 35-44 focused campaign	Creative Team
Month 1	Convert all campaigns to 30-day	Campaign Manager
Month 2	Increase Houston/Miami budget	Regional Head
Quarter 2	Review results and iterate	Analytics Team